

# Conference Presentation Outline (15 min)

Slide-by-slide plan for a 15 minute Work-in-Progress talk based on the hybrid hardware courses paper. Budget: ~12 to 13 minutes speaking, 2 to 3 minutes for questions. Arc: problem to design to early findings to next steps.

## Slide 1 - Title (~15 sec)

Title, name and affiliation (Cornell Systems Engineering), venue. One line framing it as a WIP case study so expectations are set: documents implementation and early observations, not proven efficacy.

## Slide 2 - The Problem (~1 min)

The motivating tension. Distance education roughly doubled from 18% to 45% of exclusively-online enrollment between 2019 and 2020, and the shift is structural, not just pandemic-driven (growth of professional M.Eng. programs, working professionals wanting part-time degrees). The catch: hardware-intensive labs are the hardest thing to deliver remotely because they depend on direct physical manipulation and sensory feedback. This is the "why care" slide.

## Slide 3 - Why Existing Solutions Fall Short (~1 min)

Quick contrast of alternatives. Simulations and remote-access labs exist but keep students at arm's length and create a "black box" effect. Take-home kits preserve physical debugging but introduce their own problems: loss of peer coordination, weaker oral communication, harder integrity verification. Then name the new wrinkle: generative AI can produce code and lab reports without conceptual understanding. Sets up the two interventions (kits + oral exams).

## Slide 4 - Research Questions (~45 sec)

The three questions, lightly trimmed:

1. Do standardized at-home kits achieve technical and pedagogical parity?
2. Do periodic oral assessments support engagement and integrity in the age of AI?
3. Does the flipped format increase perceived parity?

Note these emerged from building the courses, and the paper lays groundwork rather than answering definitively.

## Slide 5 - The Two Courses (~1 min)

Introduce SYSEN 5411 (Intro to Robotics) and 5412 (Creating Solutions with Embedded Systems). Both one-credit, hybrid, flipped. Identical kits shipped to distance learners. Cohort: 44 students total, 27 on-campus and 17 distance learners, the latter largely working professionals. Plant the on-campus vs. distance contrast that pays off in results.

## **Slide 6 - Design Decision 1: Flipped + Hybrid (~1 min)**

Lectures asynchronous via pre-recorded video; weekly synchronous sessions reserved for demos and troubleshooting. Sessions recorded for remote students, with Ed Discussion for peer support incentivized by participation points. Key insight: pre-recording freed synchronous time for the high-value hands-on work students said mattered most.

## **Slide 7 - Design Decision 2: Hardware + Language (~1.5 min)**

Two linked choices driven by parity. Hardware: tool-less kits requiring no soldering iron or oscilloscope, with strong online documentation (XRP for robotics, SunFounder Pico kit for embedded). Language: MicroPython, justified by the survey showing 93% Python familiarity versus far fewer with C or C++, and 43% being hardware novices. Point: reduce the syntax and tooling hurdle so cognitive load goes to systems logic. Good slide for a pie chart (majors or language familiarity).

## **Slide 8 - Assessment: Oral Exams + Workload (~1.5 min)**

The most distinctive contribution, so give it room. Oral exam format: a 20-minute Zoom design review after the planning phase, covering user interaction, system mapping, and implementation strategy. Frame as both AI-mitigation tool and relational touchpoint. Pair with the workload finding: the "two-hour rule" and the brown-out debugging story (students losing up to 7 to 9 hours to dead AA batteries), which led to making the robotics final project optional. The battery anecdote is memorable and worth telling out loud.

## **Slide 9 - Early Result: The Engagement Surprise (~1.5 min)**

The headline finding. Contrary to expectations, distance learners were more engaged and ambitious than on-campus students, treating the project as a creative break, while on-campus students under heavy full-time loads defaulted to minimum viable products. Oral exams reliably surfaced students who had skipped foundational labs (near-perfect correlation between incomplete weekly work and weak design reviews). Reframes the distance-learner-as-disadvantaged assumption.

## **Slide 10 - Early Result: AI Usage (~1 min)**

The AI usage breakdown from Lab 1: about 36% used no AI, and the largest active use (30%) was debugging, with conceptual and code-writing uses rare. Interpretation: students used AI to

get past friction points, not to replace learning, which strengthens the case for oral exams that test the "why" while AI handles the "how." Good slide for the AI usage pie chart.

## **Slide 11 - Future Work (~1 min)**

The next iteration: boilerplate code to cut low-level debugging in robotics, a tiered three-checkpoint oral exam schedule, systematic logging of friction points, classifying AI use as "augmentative" vs. "bypass," and IRB approval for demographic analysis. Keep brisk; signals the longitudinal study to come.

## **Slide 12 - Conclusion (~45 sec)**

Three takeaways: parity needs pedagogy, not just identical hardware; tool-less kits cut logistical friction; oral exams turn assessment into authentic mentorship rather than an AI-vulnerable task. Close by positioning this as the foundation for a fuller study.

## **Slide 13 - Questions / Contact**

Contact info and an invitation. Optionally hold one backup slide with the cohort or familiarity charts for detailed questions.

## **Presenting Notes**

- Slides 8 and 9 are the core. If running long, compress slides 2 and 3 rather than those.
- The brown-out battery story and the engagement-surprise finding are the two most quotable moments. Land them clearly.
- With four figures in the paper, use at most three in the talk (language/majors, AI usage, optionally the familiarity ranking) to avoid chart fatigue.